

Answer (Topic 5 Magnetism & Electromagnetic Induction)

1.

$$B_1 = \frac{\mu_0 I_1}{2\pi r} = \frac{(4\pi \times 10^{-7})(5)}{2\pi(0.03)} = 3.3335 \times 10^{-5} \text{ T}$$

$$B_1 = B_2$$

$$B_{total} = B_1 + B_2$$

$$B_{total} = 6.667 \times 10^{-5} \text{ T}$$

2.

$$B = \frac{\mu_0 N I}{l} = \frac{(4\pi \times 10^{-7})(500)(0.6)}{(0.03)} = 0.013 \text{ T}$$

3.

$$B_p = \frac{\mu_0 I_p}{2\pi r} = \frac{(4\pi \times 10^{-7})(40)}{2\pi(0.05)} = 1.6 \times 10^{-4} \text{ T}$$

$$B_Q = \frac{\mu_0 I_Q}{2\pi r} = \frac{(4\pi \times 10^{-7})(20)}{2\pi(0.05)} = 8.0 \times 10^{-5} \text{ T}$$

$$B_{total} = B_p + B_Q$$

$$B_{total} = 2.4 \times 10^{-4} \text{ T (upward)}$$

4.

$$B_1 = \frac{\mu_0 I_1}{2\pi r} = \frac{(4\pi \times 10^{-7})(4)}{2\pi(0.02)} = 4.0 \times 10^{-5} \text{ T}$$

$$B_2 = \frac{\mu_0 N I_2}{2r} = \frac{(4\pi \times 10^{-7})(1)(2)}{2(0.02)} = 6.28 \times 10^{-5} \text{ T}$$

$$B_{total} = B_1 + (-B_2)$$

$$B_{total} = -2.28 \times 10^{-5} \text{ T (into the page)}$$

5.

$$B_1 = \frac{\mu_0 I_1}{2\pi r} = \frac{(4\pi \times 10^{-7})(8)}{2\pi(0.08)} = 2 \times 10^{-5} \text{ T}$$

$$B_2 = \frac{\mu_0 N I_2}{2r} = \frac{(4\pi \times 10^{-7})(1)(5)}{2(0.12)} = 2.62 \times 10^{-5} \text{ T}$$

$$B_{total} = (-B_1) + (B_2)$$

$$B_{total} = 0.62 \times 10^{-5} \text{ T (out of the page)}$$

6. Circumference of circle,

$$l = 2\pi r$$

$$2 = 2\pi r$$

$$r = \frac{1}{\pi} = 0.318 \text{ m}$$

$$B = \frac{\mu_0 N I}{2r} = \frac{(4\pi \times 10^{-7})(5)(30 \times 10^{-3})}{2(0.318)}$$

$$B = 6.83 \times 10^{-7} \text{ T}$$

- 7.

$$A = \pi r^2$$

$$\emptyset = BA \cos \theta$$

$$\emptyset = B(\pi r^2) \cos \theta$$

$$\emptyset = (0.4)(\pi 2^2) \cos 0^\circ$$

$$\emptyset = 5.03 \text{ Wb}$$

- 8.

(a) $\emptyset_N = NBA \cos \theta$

$$\emptyset_N = (50)(2.8)(4.5 \times 10^{-4}) \cos 0^\circ$$

$$\emptyset_N = 0.063 \text{ Wb}$$

(b) $\mathcal{E}_{ind} = -NA \frac{dB}{dt}$

$$\mathcal{E}_{ind} = -(50)(4.5 \times 10^{-4}) \frac{(0 - 2.8)}{(0.2)}$$

$$\mathcal{E}_{ind} = 0.315 \text{ V}$$

- 9.

(a) $\mathcal{E}_{ind} = Blv \sin \theta$

$$\mathcal{E}_{ind} = (1.5)(0.4)(0.02) \sin 90^\circ$$

$$\mathcal{E}_{ind} = 0.012 \text{ V}$$

$$\begin{aligned} \text{(b)} \quad \mathcal{E}_{ind} &= I_{ind} R \\ 0.012 &= I_{ind} (3) \\ I_{ind} &= 4 \times 10^{-3} \text{ A} \end{aligned}$$

10.

$$\begin{aligned} \text{(a)} \quad \mathcal{E}_{ind} &= Blv \sin \theta \\ \mathcal{E}_{ind} &= (4)(0.2)(0.5) \sin 90^\circ \\ \mathcal{E}_{ind} &= 0.4 \text{ V} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \mathcal{E}_{ind} &= I_{ind} R \\ 0.4 &= I_{ind} (10) \\ I_{ind} &= 0.04 \text{ A} \end{aligned}$$